Project #4: S-To-Z Transformations and S-domain Modeling of SC Circuits

41. Construct the filter using the behavioral (VerilogA) adder, scaler, and integrator in the EEE598Lib and verify your results using Spectre AC simulation. (15 points)

Shown in Figure 1 is the design of the 5th order Butterworth Filter using adders, scalers, and integrators. The design is organized into three stages. The first stage features one integrator (1st order) and a feedback loop of +1. A Q value of 0.62 is used for the second and third stages of the filter. The second stage features two integrators (3rd order) with a G^2 feedback of -1 and a G feedback of 1/Q. The third stage also features two integrators (5th order) with a G^2 feedback of -1 and a G feedback of Q. This design ensures proper functionality of the 5th Order Butterworth filter whose AC response is shown in Figure 2.

Figure 1: Schematic of 5th Order Butterworth Filter with Adders, Scalers, and Integrators

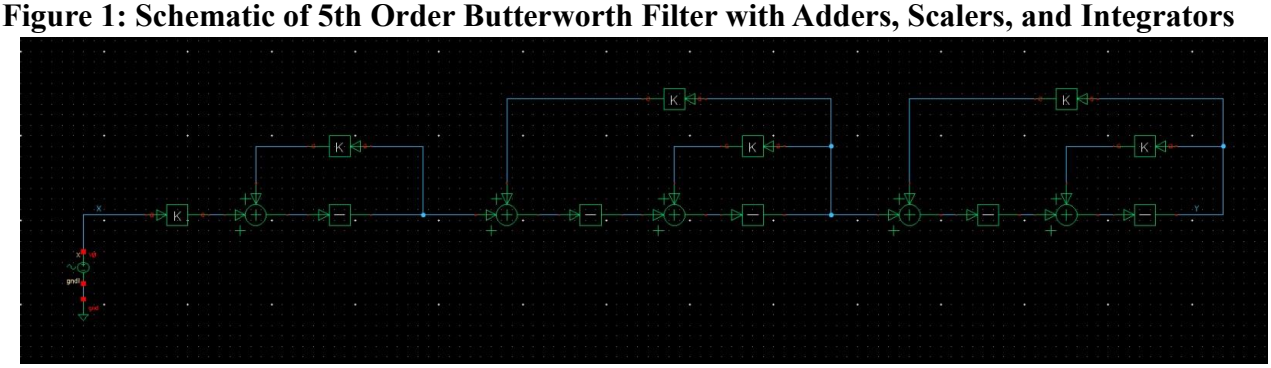
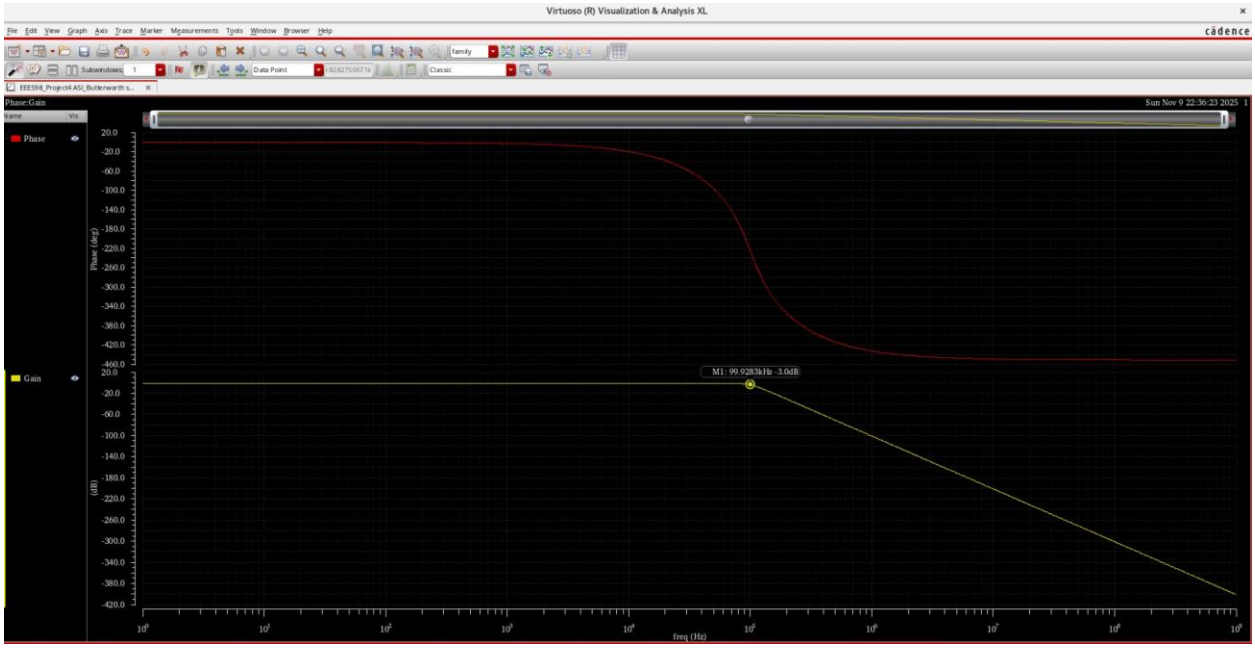


Figure 2: AC Response of 5th Order Butterworth Filter with Adders, Scalers, and Integrators



-3dB Frequency = 99.93 kHz

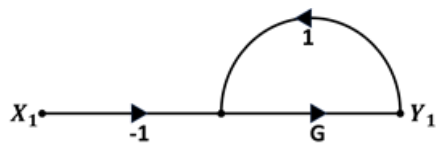
4.2. Construct the filter using the Active-RC circuit (using ideal OP/Cap/resistor) and verify your results using Spectre (or SPICE) AC simulation (15 points) Transfer function for 5th order lowpass Butterworth filter is:

$$H(s) = \frac{1}{(s+1)(s^2+1.618s+1)(s^2+0.618s+1)}$$

Let $H(s) = H_1(s) \times H_2(s) \times H_3(s)$ **Where**
$H_1(s) = \frac{1}{s+1}$,$H_2(s) = \frac{1}{s^2+1.618s+1}$ **and** $H_3(s) = \frac{1}{s^2+0.618s+1}$ **To implement using VerilogA components, Active-RC and Gm-C methods, using** $G = -\frac{1}{s}$ **Replace** s $\left(-\frac{1}{G}\right)$ **with**

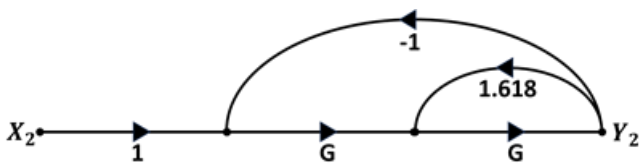
$$H_1(s) = \frac{Y_1}{X_1} = \frac{1}{-\frac{1}{G}+1} \rightarrow Y_1 = (Y_1 - X_1)G$$

SFG for $H_1(s)$ **is:**



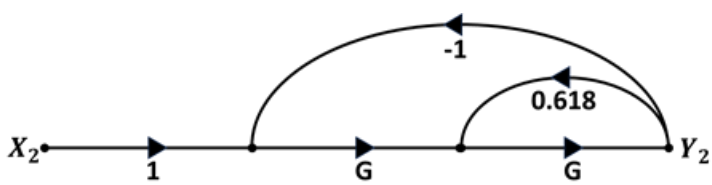
$$H_2(s) = \frac{Y_2}{X_2} = \frac{1}{\frac{1}{G^2} - \frac{1.618}{G} + 1} \rightarrow Y_2 = (X_2 - Y_2)G^2 + 1.618GY_2$$

SFG for $H_2(s)$ is:



$$H_3(s) = \frac{Y_3}{X_3} = \frac{1}{\frac{1}{G^2} - \frac{0.618}{G} + 1} \rightarrow Y_3 = (X_3 - Y_3)G^2 + 0.618GY_3$$

SFG for $H_3(s)$ is:



An Active-RC filter was designed using ideal op-amps, resistors, and capacitors. With R = 100 kΩ, the calculated C = 15.9 pF achieved the same 100 kHz corner. The Spectre AC plot confirmed the –3 dB point and proper phase response consistent with Butterworth behavior.

Frequency scaling, replace *s* with *(s / ω_o)*, where
$\omega_o = 2\pi(100K)$ rad/s
Choosing **R = 100 kΩ**, we get
C = 1 / (2πfR) = 1 / [2π(100K)(100K)] = 15.9 pF

Figure 3: Schematic of 5th Order Butterworth Filter with Active RC Circuit

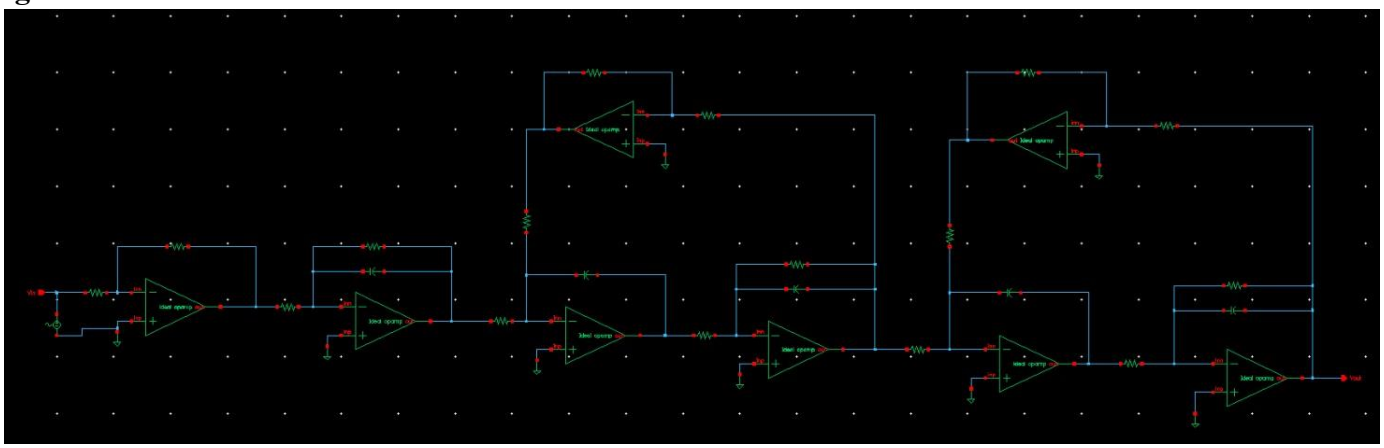


Figure 4: Ideal Op Amp Used in Active RC Circuit

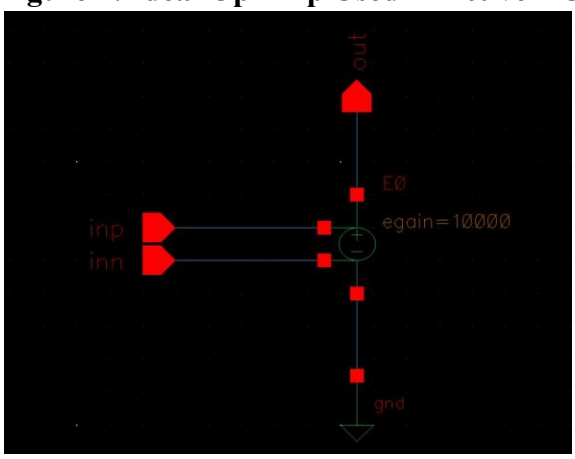
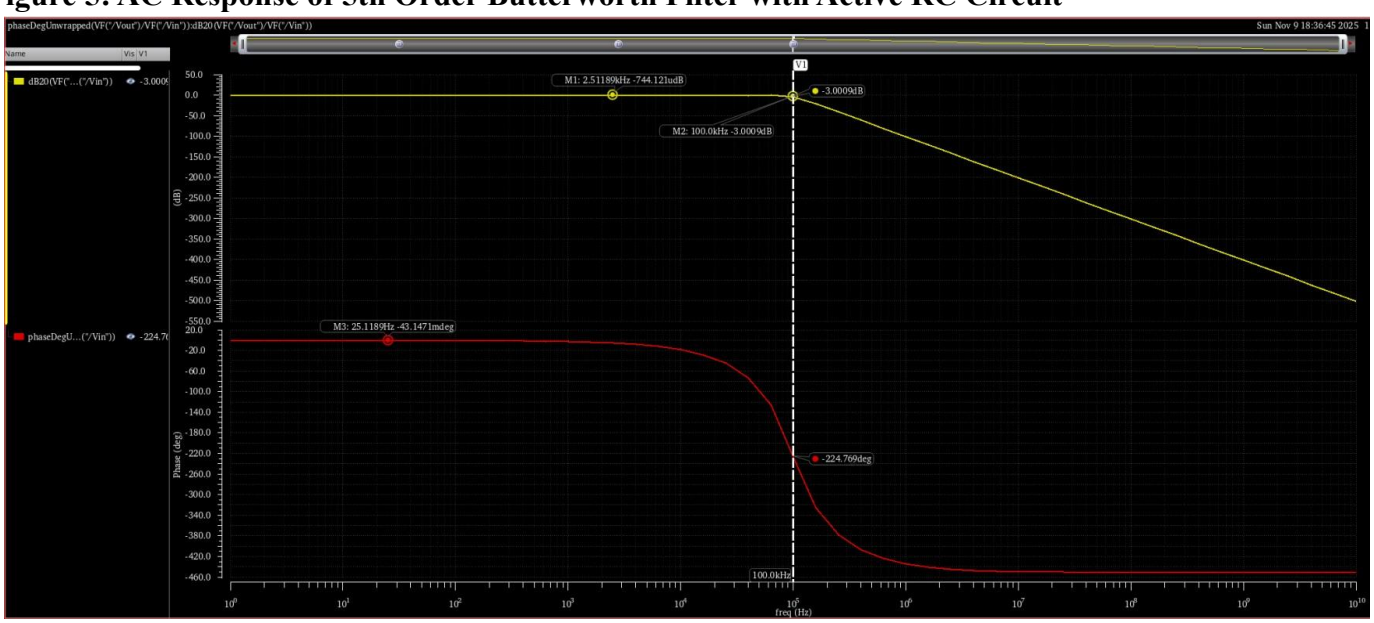


Figure 5: AC Response of 5th Order Butterworth Filter with Active RC Circuit



-3dB Frequency = 100kHz

4.3. Construct the filter using the Gm-C circuit (using ideal Gm/Cap) and verify your results using Spectre (or SPICE) AC simulation. (15 points)

Using ideal transconductors and capacitors, the Gm-C version of the filter was constructed. Setting R = 100 kΩ gave g = 1/R = 10 μS and C = 15.9 pF. AC simulation showed correct low-pass response and phase characteristics, validating the transconductance-based realization of the same transfer function.

Figure 6: Schematic of 5th Order Butterworth Filter Using Gm-C Circuit

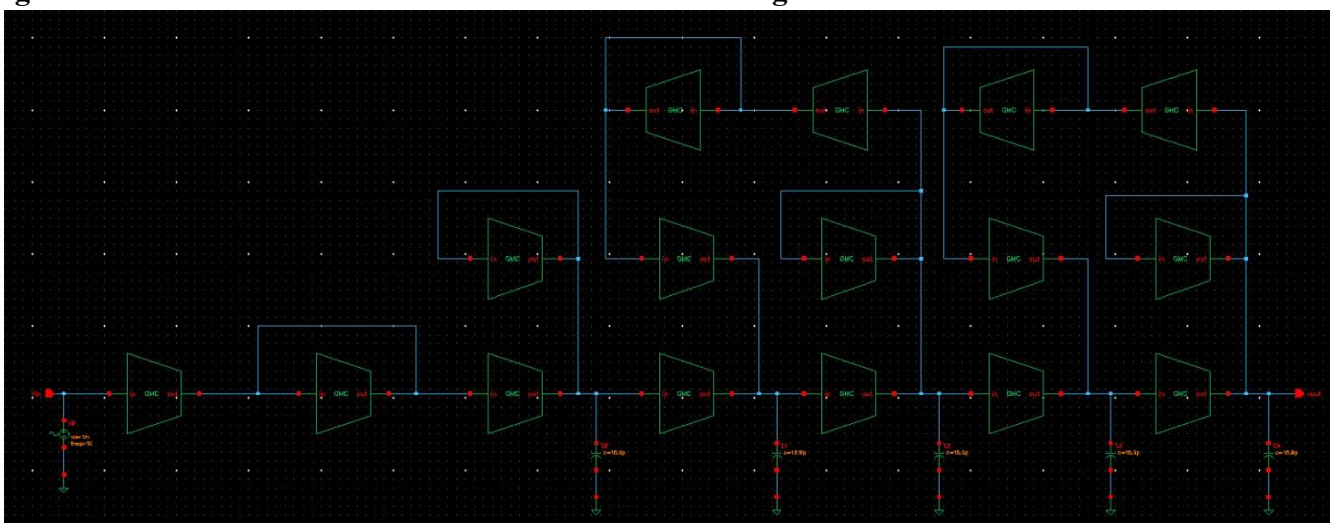
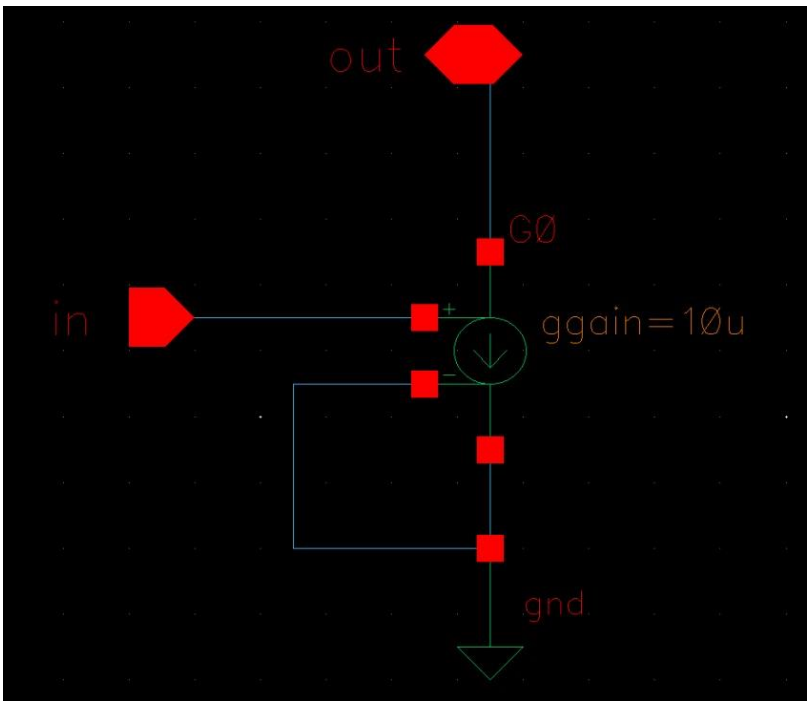


Figure 7: Ideal Gm Used

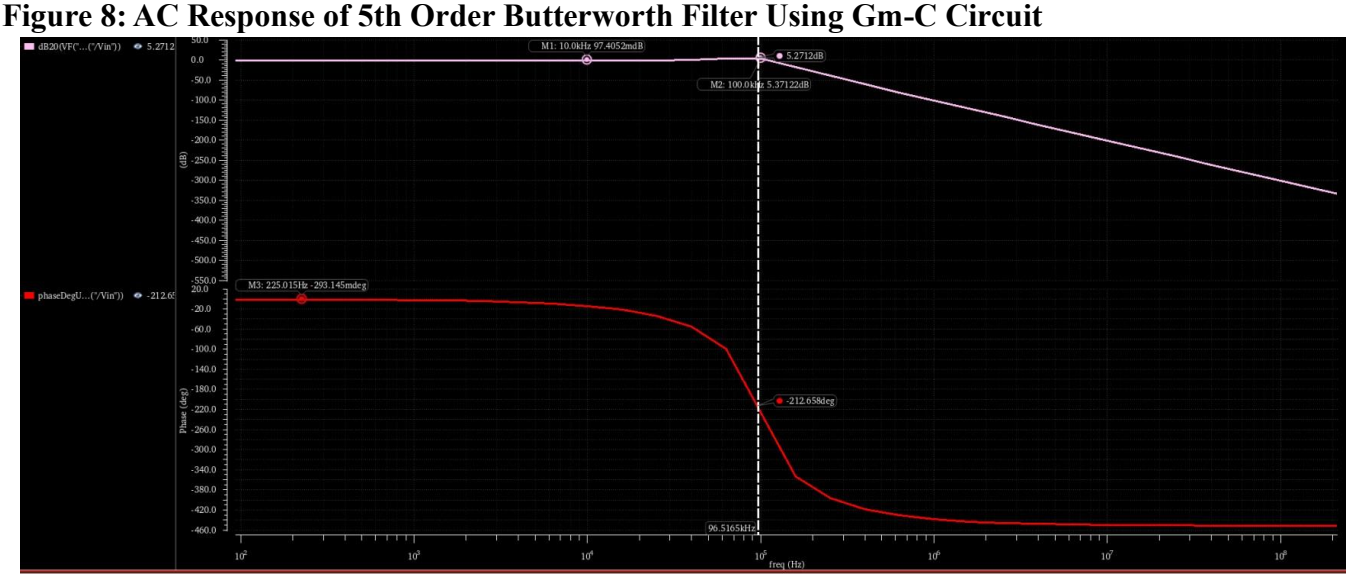


Frequency scaling, replace *s* with *(s / ω_o)*, where
$\omega_o = 2\pi(100K)$ rad/s

Choosing R = 100 kΩ, we get g = 1 / R = 10 µS and C = 1 /

$(2\pi fR) = 1 / [2\pi(100K)(100K)] = 15.9$ pF

Figure 8: AC Response of 5th Order Butterworth Filter Using Gm-C Circuit

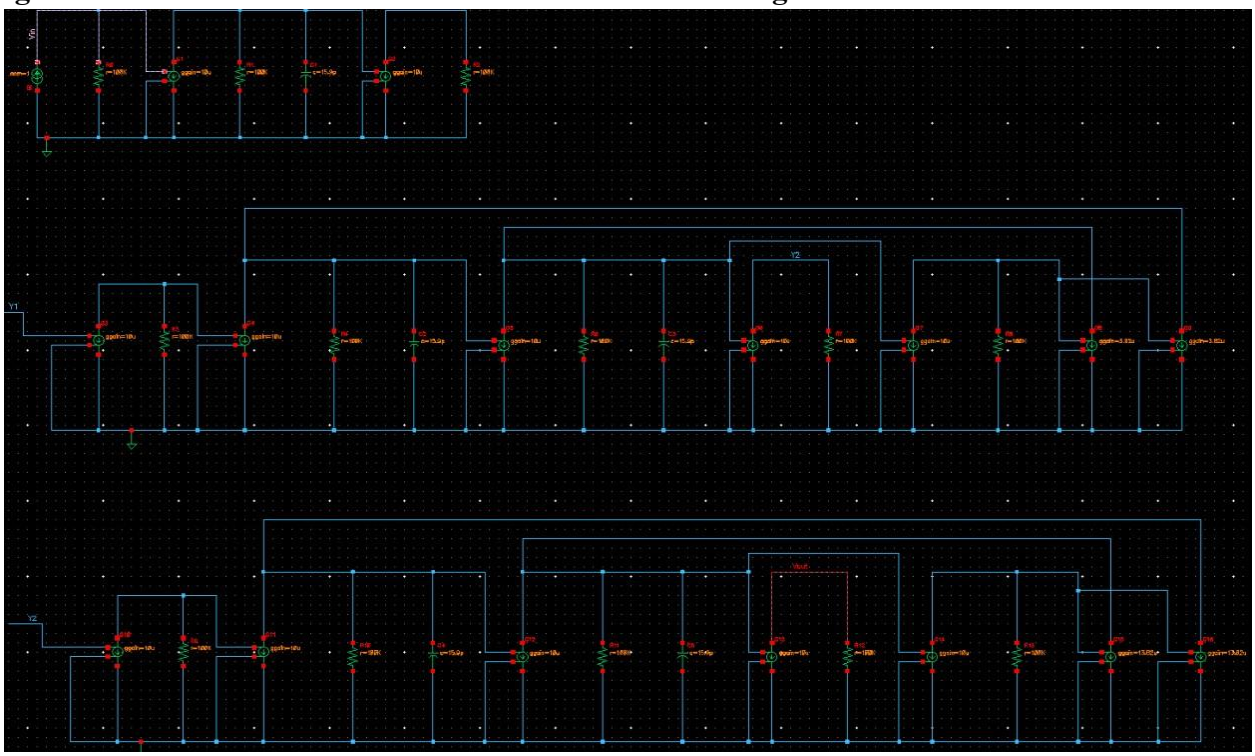


-3dB Frequency = 98.75kHz

4.4. Construct the filter using the CTI circuit (using ideal current mirror and Lossy integrator) and verify your results using Spectre (or SPICE) AC simulation (15 points) The CTI filter was realized using ideal current mirrors and lossy integrators, with g = 10 µS, R = 100 kΩ, and C = 15.9 pF. Frequency scaling was applied for $\omega_0 = 2\pi\times100$ kHz. Simulation verified stable current-mode operation with comparable magnitude and phase performance to previous implementations.

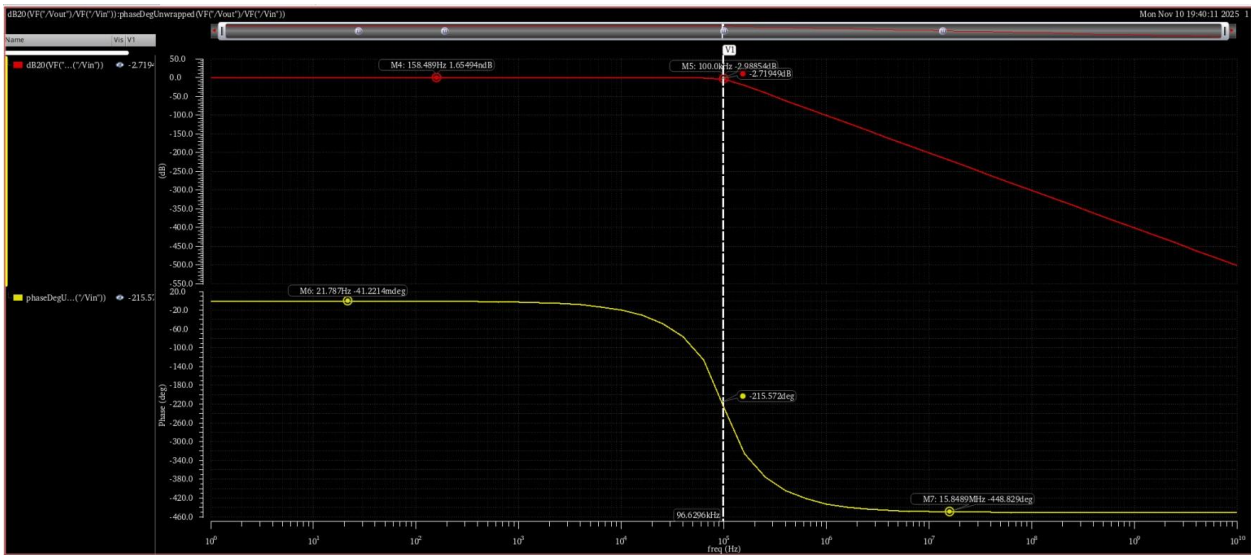
Frequency scaling, replace *s* with (s / ω_0), where $\boldsymbol{\omega_0 = 2\pi(100K)}$ rad/s.
Choosing **R = 100K** and **g = 10 µS**, and **C = 15.9 pF**.

Figure 9: Schematic of 5th Order Butterworth Filter Using CTI Circuit



Figure

10: AC Response of 5th Order Butterworth Filter with Active RC Circuit



-3dB Frequency = 100kHz

4.5. Design the filter using the Active-RC circuit (using ideal Cap/resistor and the Opamp design from project#1) and verify your results using Spectre (or SPICE) AC and TRAN simulation (40 points) the filter was rebuilt using the custom op-amp designed in Project #2. The amplifier provided a DC gain of 71 MHz, unity-gain bandwidth of 137 MHz, and a phase margin of 67°. With C = 1 pF, the calculated R = 1.59 MΩ ensured a 100 kHz cutoff. The AC and transient results confirmed correct low-pass response and good stability, though higher-frequency poles (~6 MHz) began to affect the gain and phase.

Figure 11: Cadence Circuit Schematic for Opamp from Project 2:

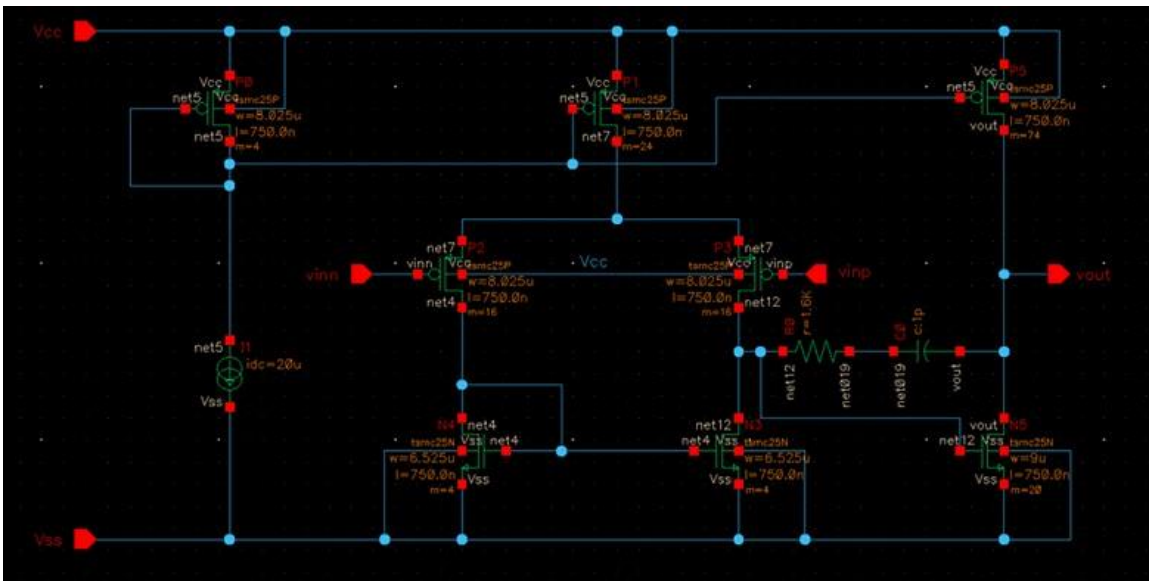

Vcc
Vss
vinn
vinp
vout
net5
net7
net4
net12
net019
I1
idc=20u
P2
P3
P1
P5
N4
N3
N5
w=8.025u
l=750.0n
w=6.525u
w=9u
r=1.6K
c:1p

Figure 12: Opamp Symbol to be Used for Testing

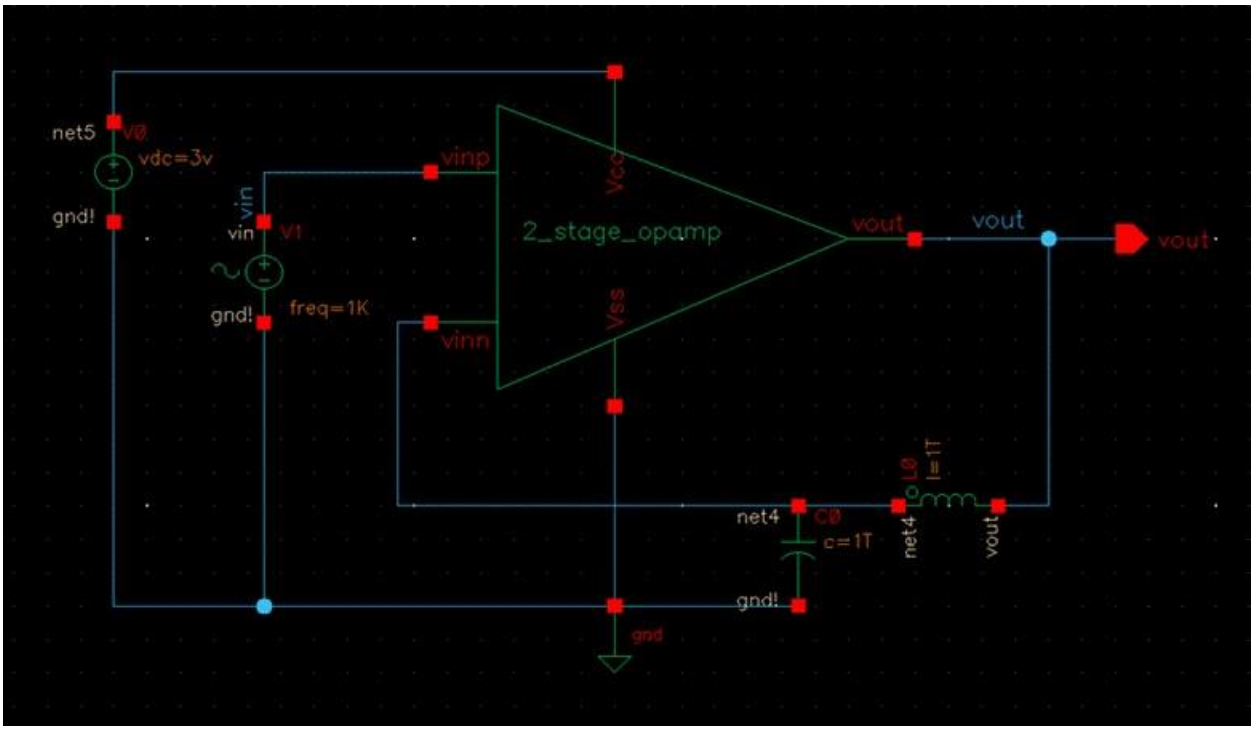
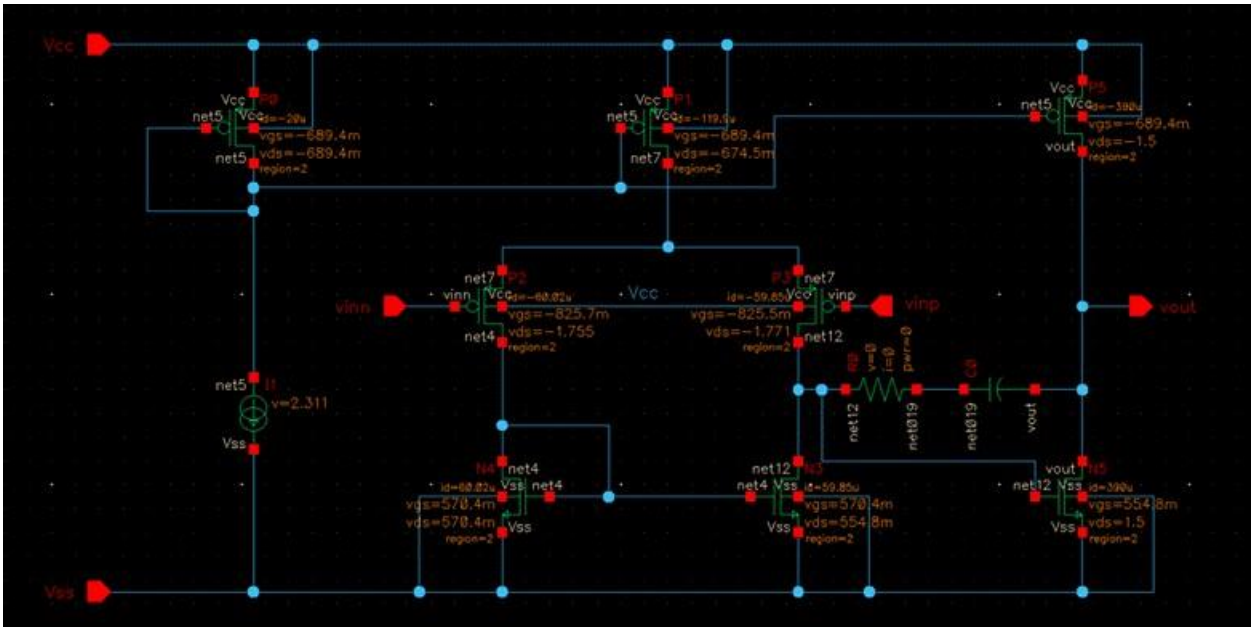
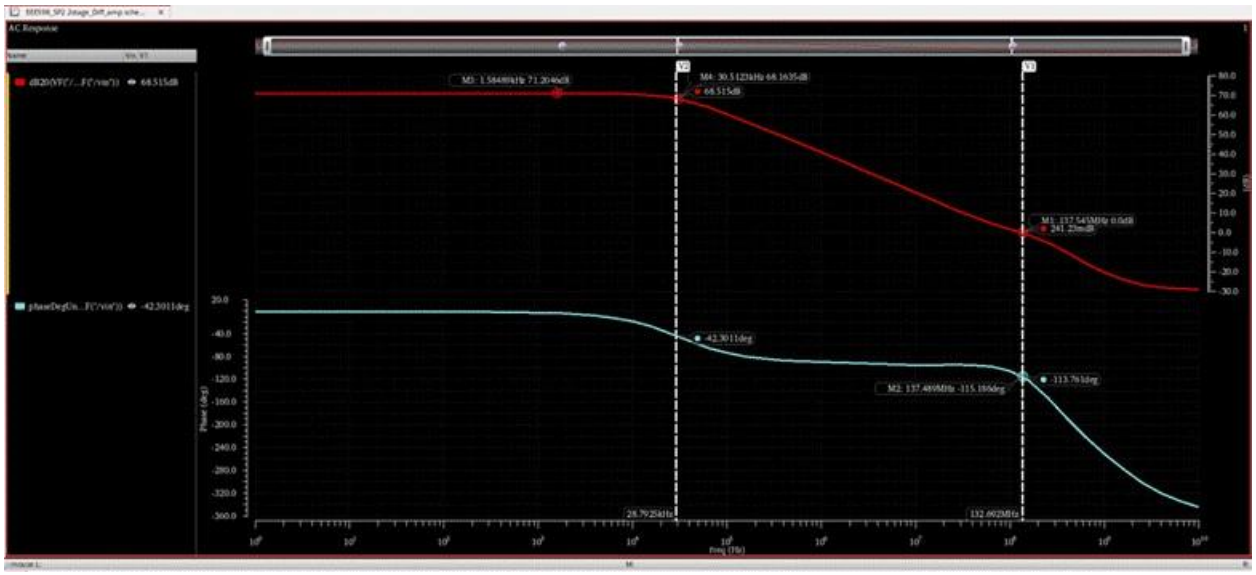


Figure 13: Two Devices In Saturation region



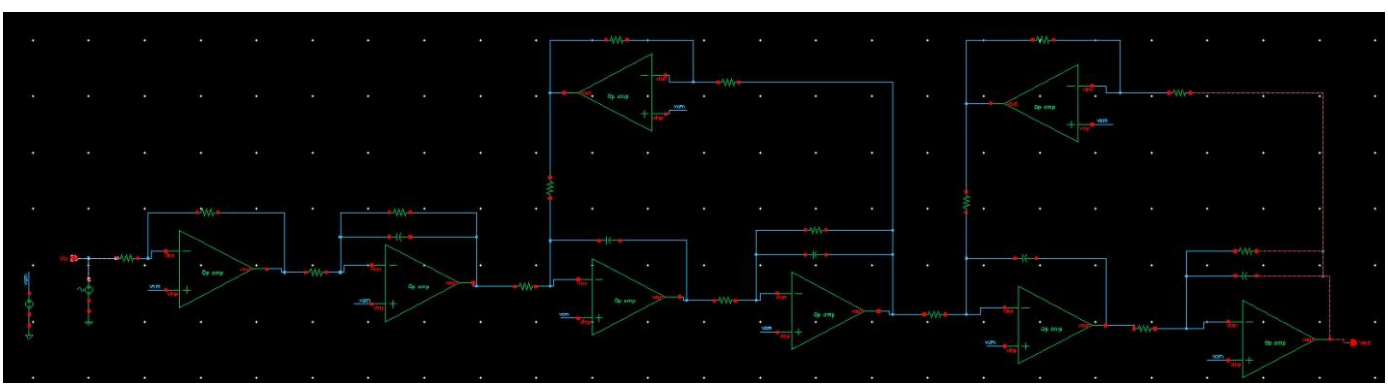
14: AC Response of Opamp Circuit



Dc gain = 71dB, unity gain frequency =137Mhz, phase margin = 67deg

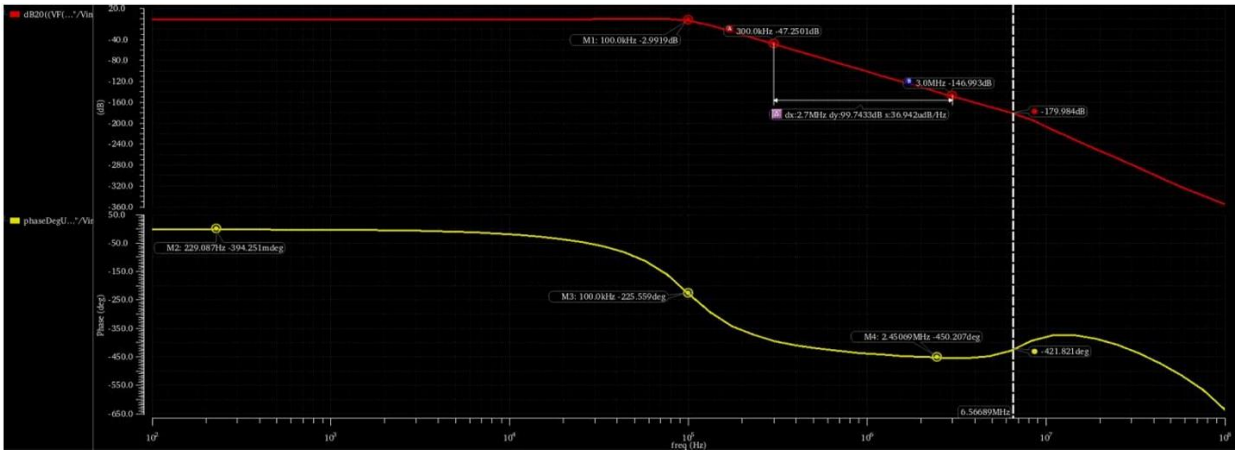
Frequency scaling, replace *s* with *(s / ω_o)*, where $\omega_o = 2\pi(100K)$ rad/s.
For stability, choosing C = 1 pF, we get
$R = 1 / (2\pi fC) = 1 / [2\pi(100K)(1p)] = 1.59\ M\Omega$
The simulation is run for VDD = 3V and Vcm = 1.5V

Figure 15: Schematic of 5th Order Butterworth Filter Using Active RC and Opamp from Project 2



Figure

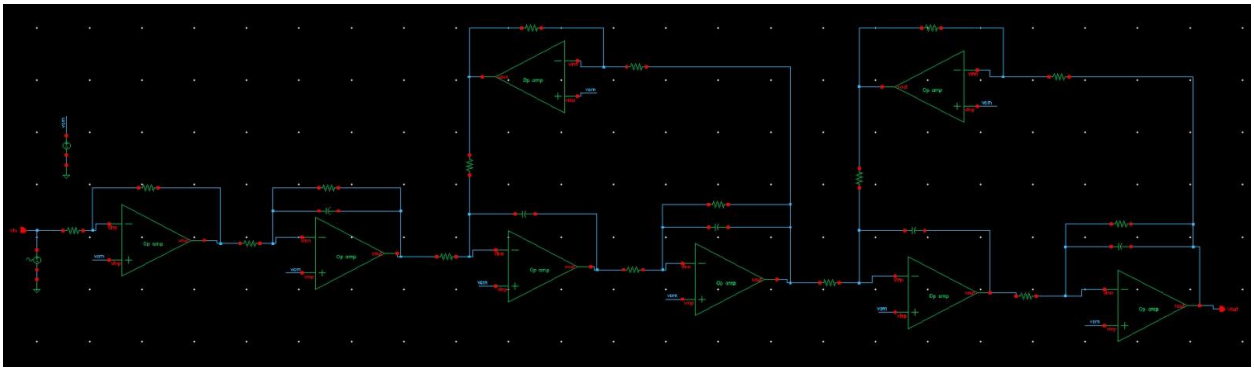
Figure 16: AC Response of 5th Order Butterworth Filter Using Active RC and Opamp from Project 2



-3dB Frequency = 100kHz

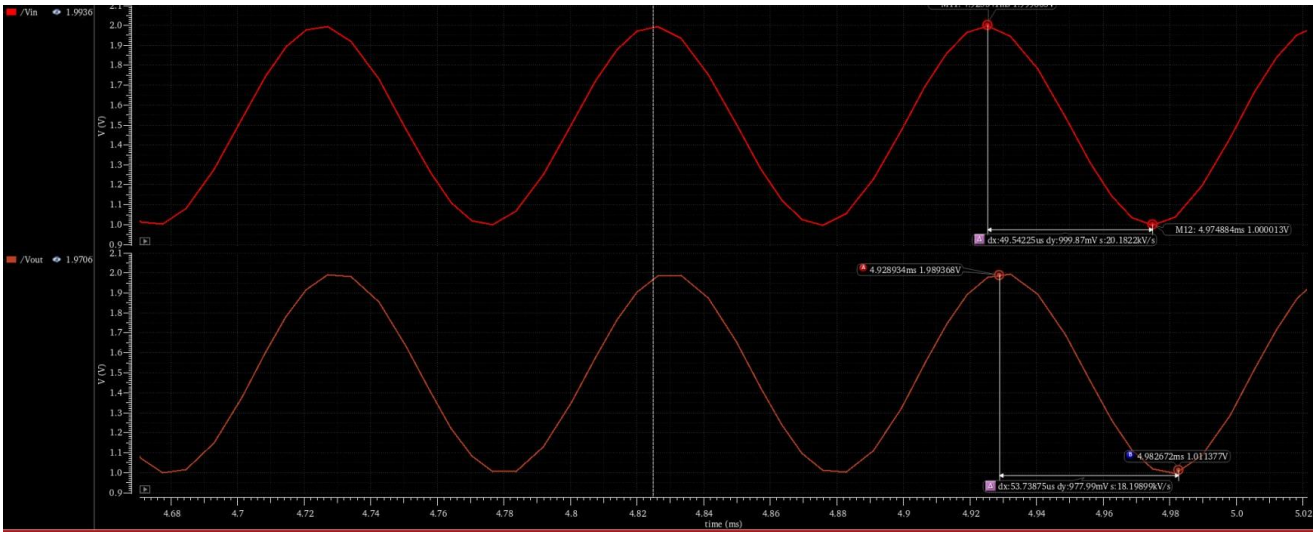
Transient analysis

Figure 17: Transient analysis of 5th Order Butterworth Filter Using Active RC and Opamp from Project 2



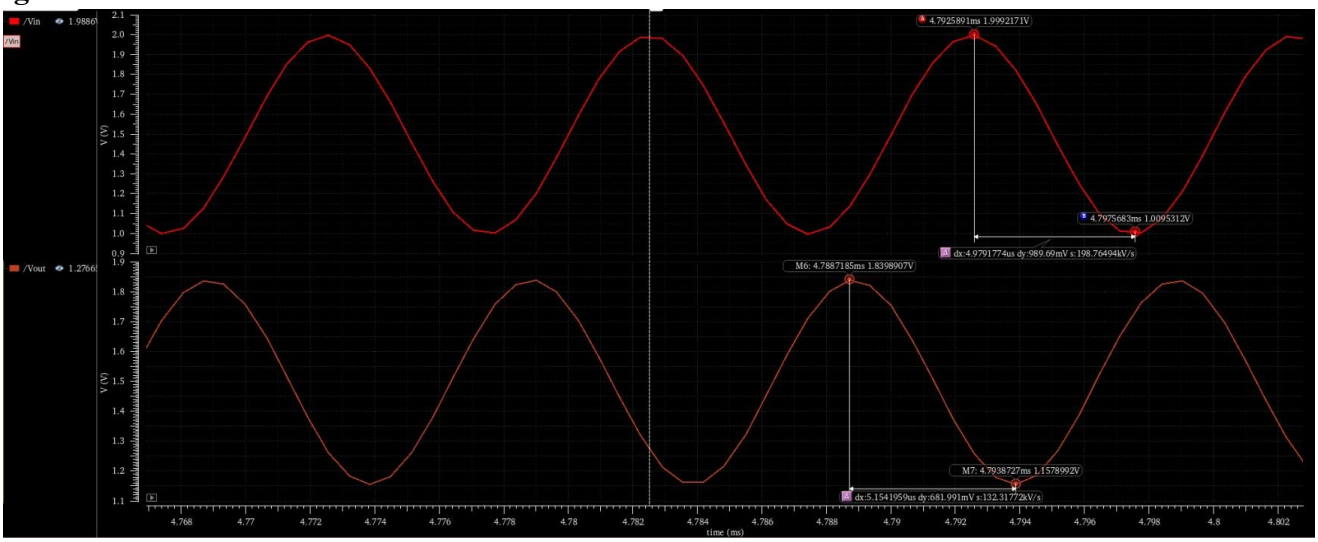
Dc voltage of 1.5V, with amplitude of 0.5V with frequency 10k, 100k and 1M are given at the input and output is observed.

18: fin at 10kHz



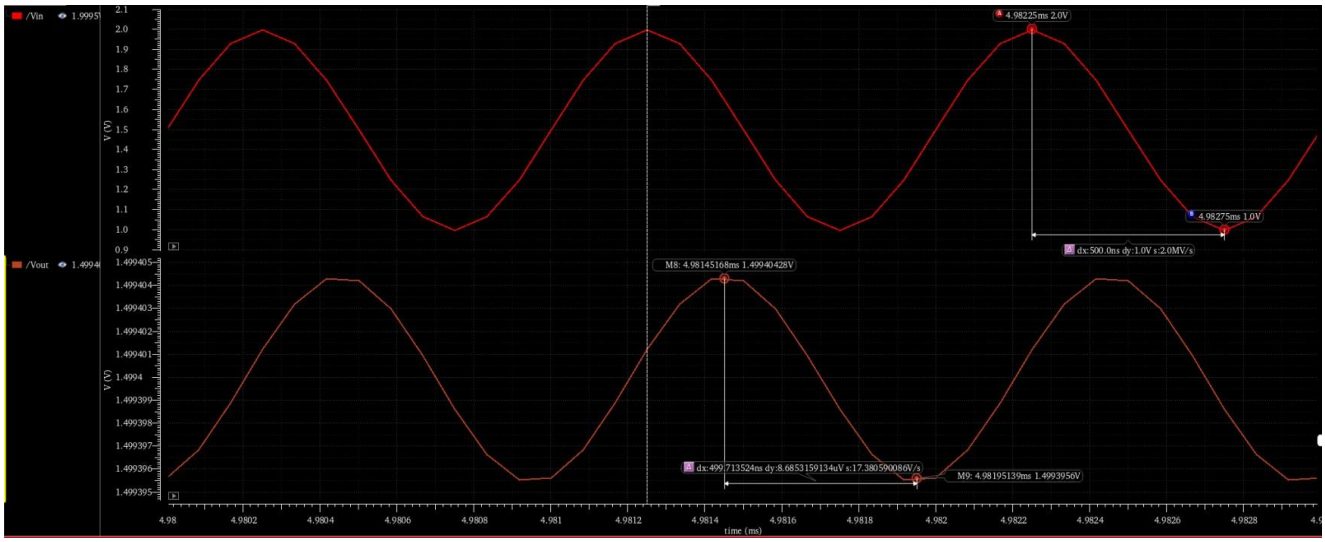
Vout ~ Vin as the dc gain here is 0db (1v/v)
dVin =1V dVout=977.99mV (~1)
Phase shift ~ 0deg

Figure 19: fin at 100kHz



Vout =0.7 x Vin as the dc gain here is -3dB
dVin =1V dVout=681mV (~0.7)
Phase ~45deg

Figure
Figure 20: fin at 1MHz



Vout =(1/100000) x Vin as the dc gain here is ~(-100dB)
dVin =1V dVout=8.68uV Phase ~ 90deg